



CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE | DELHI NCR | PUNE

AI-Powered Credit Risk Intelligence Platform

Explainable AI for Credit Default Risk Assessment

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Mission

Christ University is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment

Vision

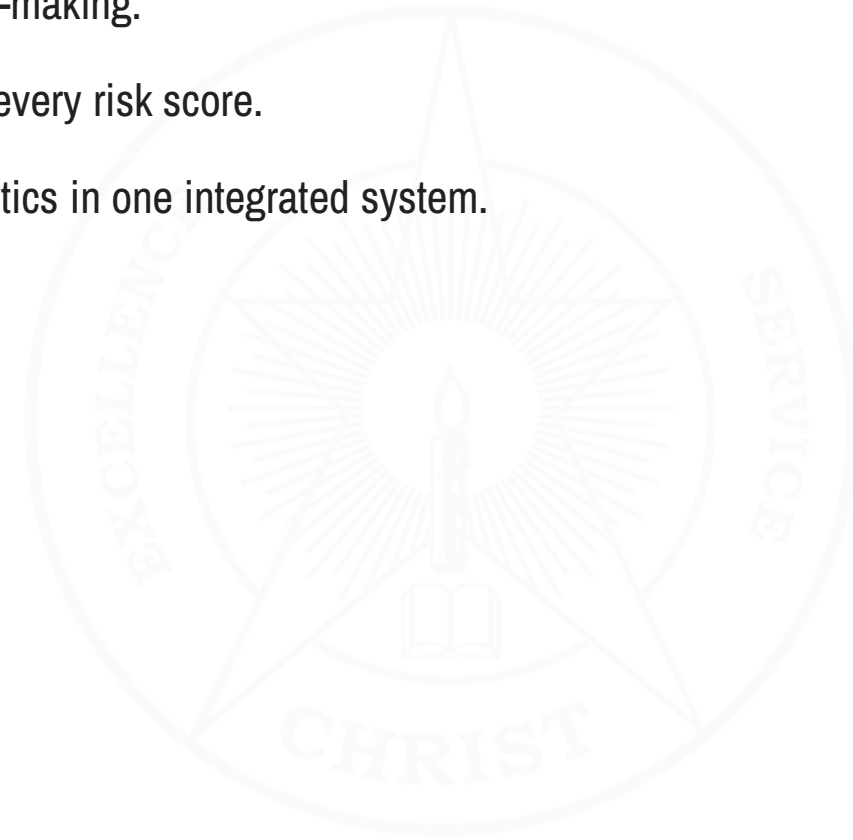
Excellence and Service

Core Values

Faith in God | Moral Uprightness
Love of Fellow Beings | Social Responsibility
Pursuit of Excellence

Business Problem

- **Identifying risk:** Financial institutions need to identify applicants at higher risk of loan default before approval.
- **Beyond prediction:** A default-probability score alone is not sufficient for confident decision-making.
- **Need for evidence:** Credit analysts require explanations and data-driven evidence behind every risk score.
- **Our solution:** This platform provides risk scoring, explainability and natural-language analytics in one integrated system.



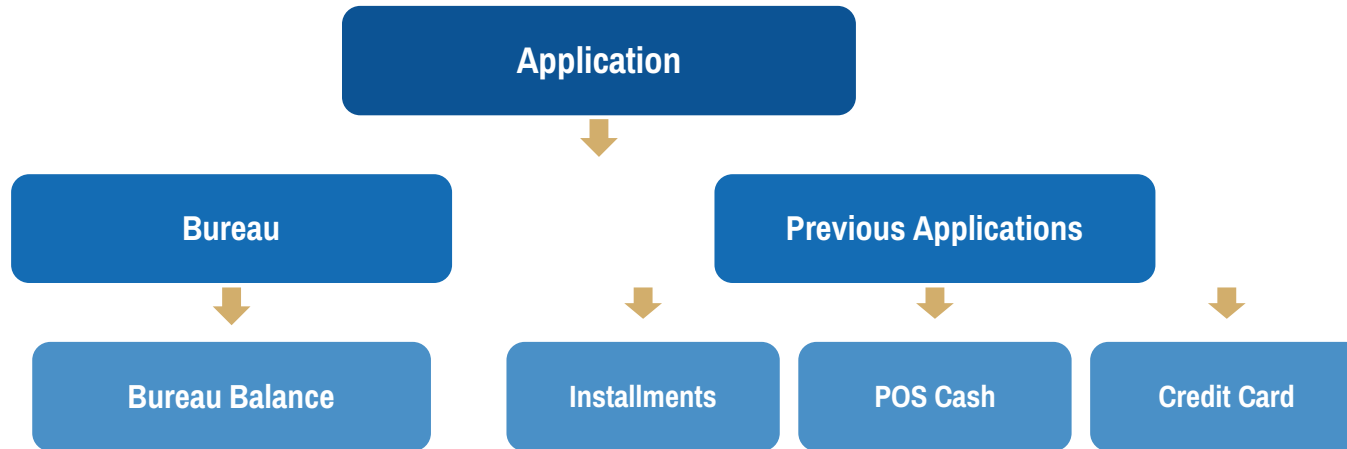
Use Case & Objectives



Platform Objectives

- Predict probability of loan default for each applicant
- Classify applicants into clear risk categories (bands)
- Provide explainable, transparent predictions (not a black box)
- Apply business rules for decision-support guidance
- Enable natural-language, conversational data analytics (Talk-to-Data)

Data Engineering & Feature Engineering

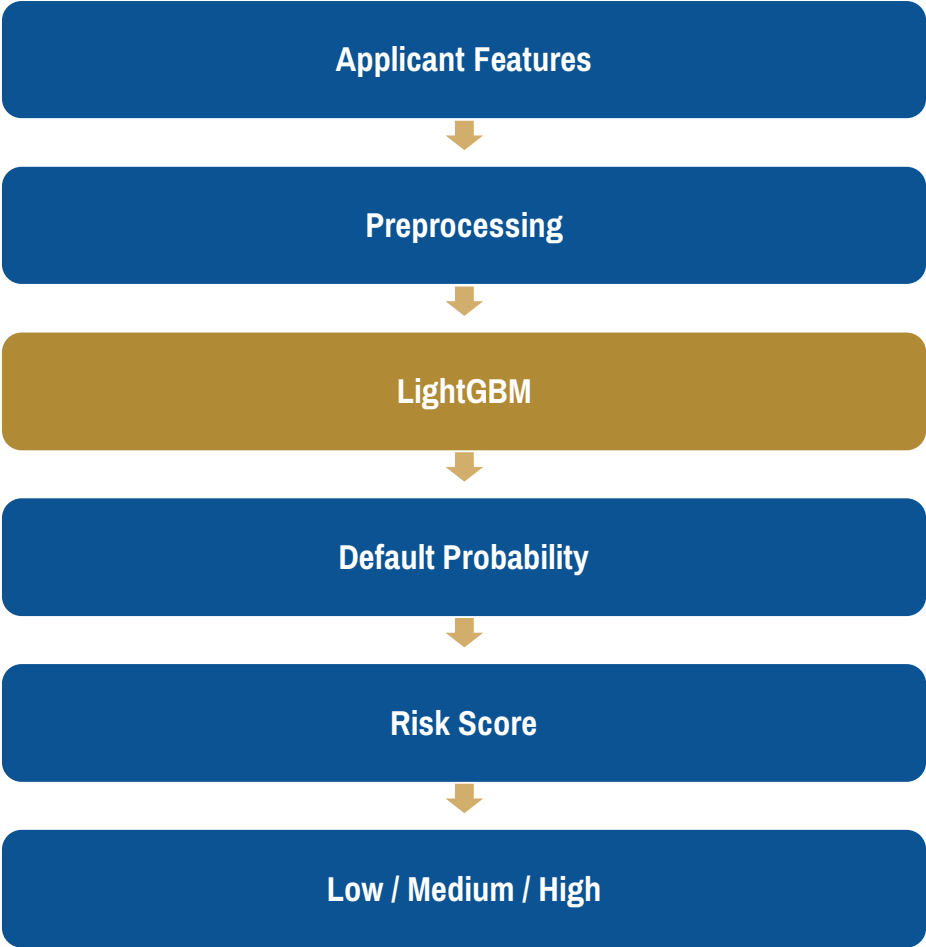


Feature Engineering Pipeline

- Aggregated historical bureau & prior-application records
- Engineered ratio, trend and behavioural features
- Merged multi-table Home Credit dataset into one feature matrix

Final Dataset: 164 Feature Columns

Machine Learning Model



Model Performance

Metric	Result
ROC-AUC	0.7762
Average Precision	0.2727
Recall	65.48%
Precision	19.19%
F1 Score	29.69%

Class imbalance: 91.93% non-default vs 8.07% default

Risk Classification

LOW
Default Probability < 30%

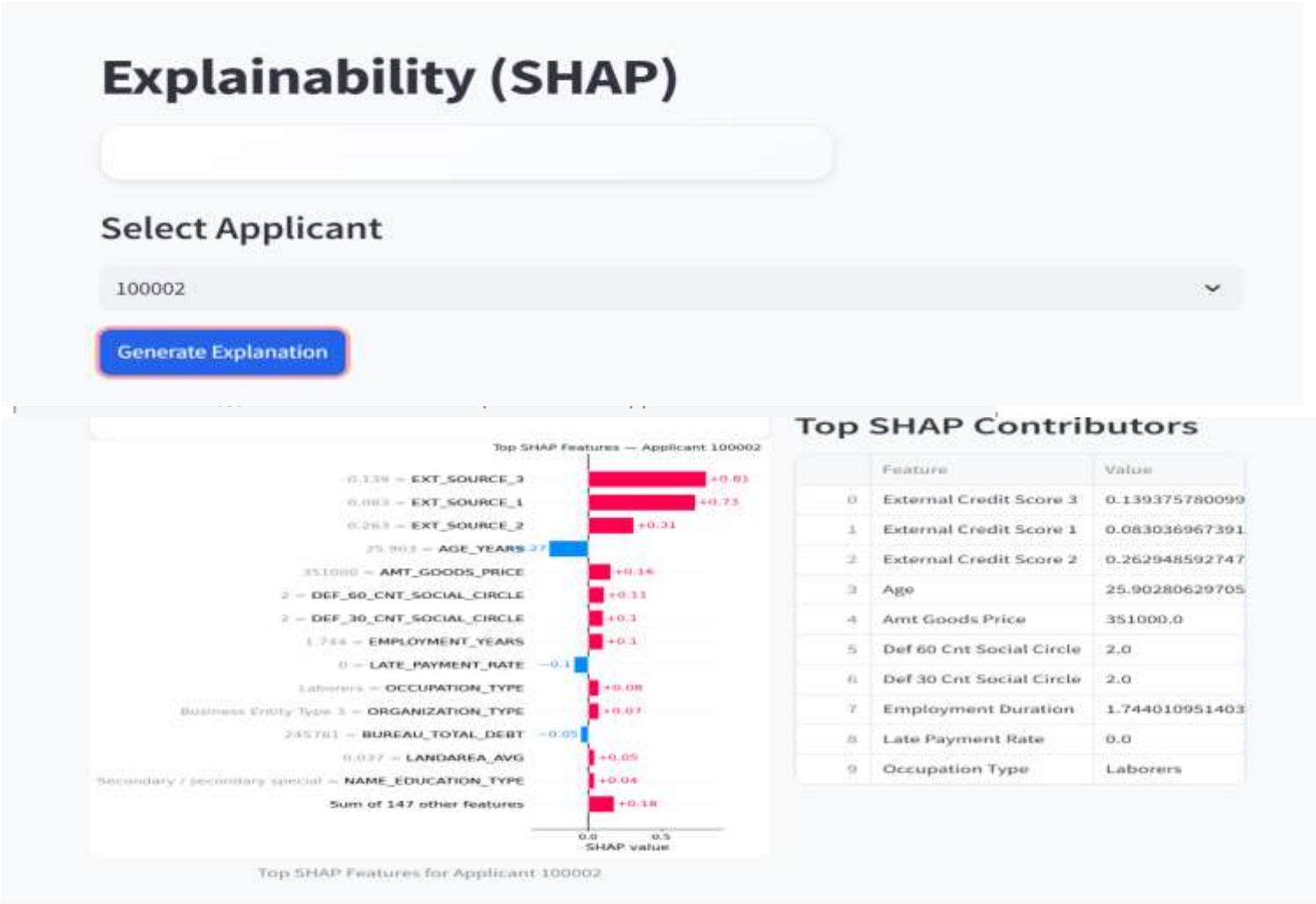
MEDIUM
Default Probability 30% – 60%

HIGH
Default Probability ≥ 60%

Risk Band	Applicants	Observed Default Rate
Low	28,085	2.44%
Medium	22,504	7.61%
High	10,914	23.51%

Risk bands show clear separation — observed default rate rises consistently from Low to High.

Explainable AI (SHAP)



Applicant 100002

Risk Level

HIGH

Predicted Default Probability

84.12%

Top Contributing Factors

- Factors increasing risk (from SHAP values)
- Factors decreasing risk (from SHAP values)

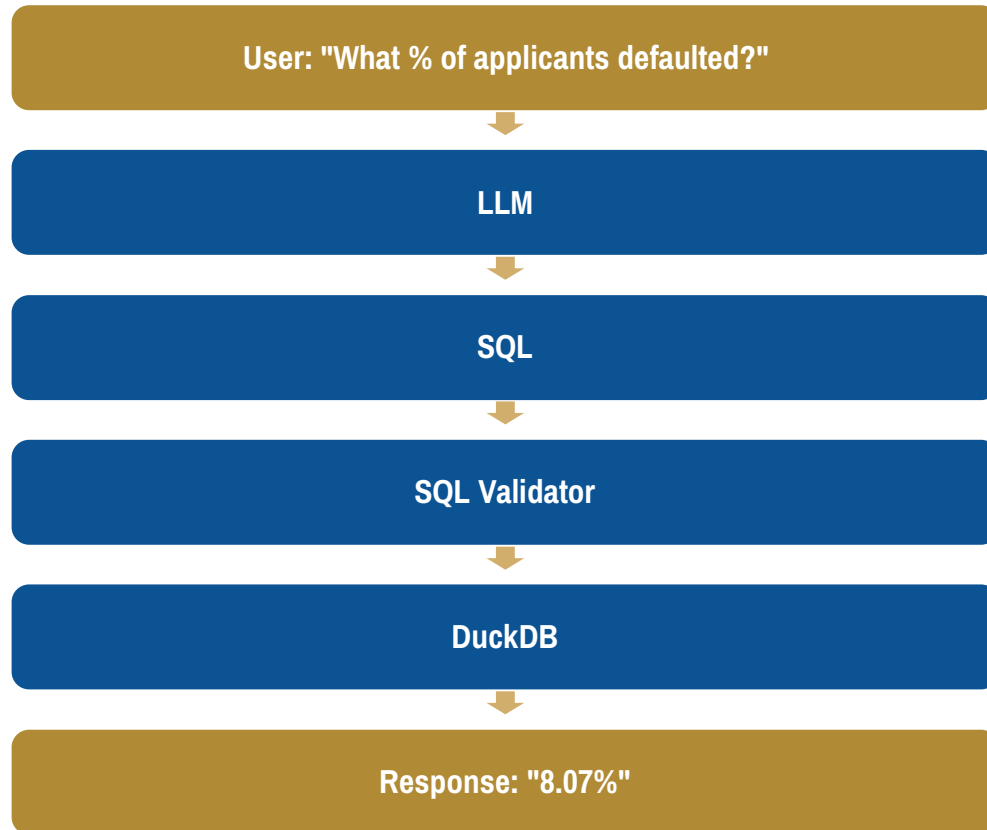
Business Risk Rules

- External credit score patterns
- Employment duration
- Late payment behaviour
- Historical credit behaviour

These are decision-support insights, not autonomous approval / rejection rules.



Talk-to-Data (Conversational Analytics)



Key Safeguards

- SQL validation before execution
- Hallucination control
- Schema grounding to the underlying dataset

Talk to Your Credit Data

Ask questions about the credit portfolio in plain English.

What is the default rate by education?

Default Rate by Education Level

The table below shows the historical default rates observed in the dataset, segmented by education type:

Education Level	Applicants	Defaults	Default Rate

Ask a question (e.g., "What is the default rate by education type?")

Platform User Interface

CREDIT RISK
INTELLIGENCE

Navigation

- Overview
- EDA & Portfolio Insights
- Risk Prediction
- Explainability
- Business Rules
- Talk to Data

Dataset

Home Credit Default Risk

Applicants

Credit Risk Intelligence Platform

An explainable AI platform for assessing historical credit risk patterns, predicting default probability, understanding model decisions, and exploring credit data using natural language.

Portfolio Metrics

TOTAL APPLICANTS

307,511

HISTORICAL DEFAULT RATE

8.07%

MODEL ROC-AUC

0.7762

HIGH-RISK APPLICANTS

10,914

Risk Band Summary

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INTELLIGENCE

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Dataset

Home Credit Default Risk

Applicants

Risk Prediction

Select Applicant

Applicant ID

100002

Calculate Risk Prediction

Prediction Output

Risk Band

HIGH

Default Probability

84.12%

Risk Score

84.12 / 100

Risk Explanation:

- Low: Probability below 20%
- Medium: Probability 20% - 54.99%
- High: Probability 60% or higher

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INTELLIGENCE

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Home Credit Default Risk

Applicants

Explainability (SHAP)

Select Applicant

100002

Generate Explanation

Top SHAP Contributors

Feature

Value

1

External Credit Score 3

0.133375780096

2

External Credit Score 1

0.083038967393

3

External Credit Score 2

0.262948332747

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INTELLIGENCE

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Home Credit Default Risk

Applicants

Default Rate by Education Level

What is the default rate by education?

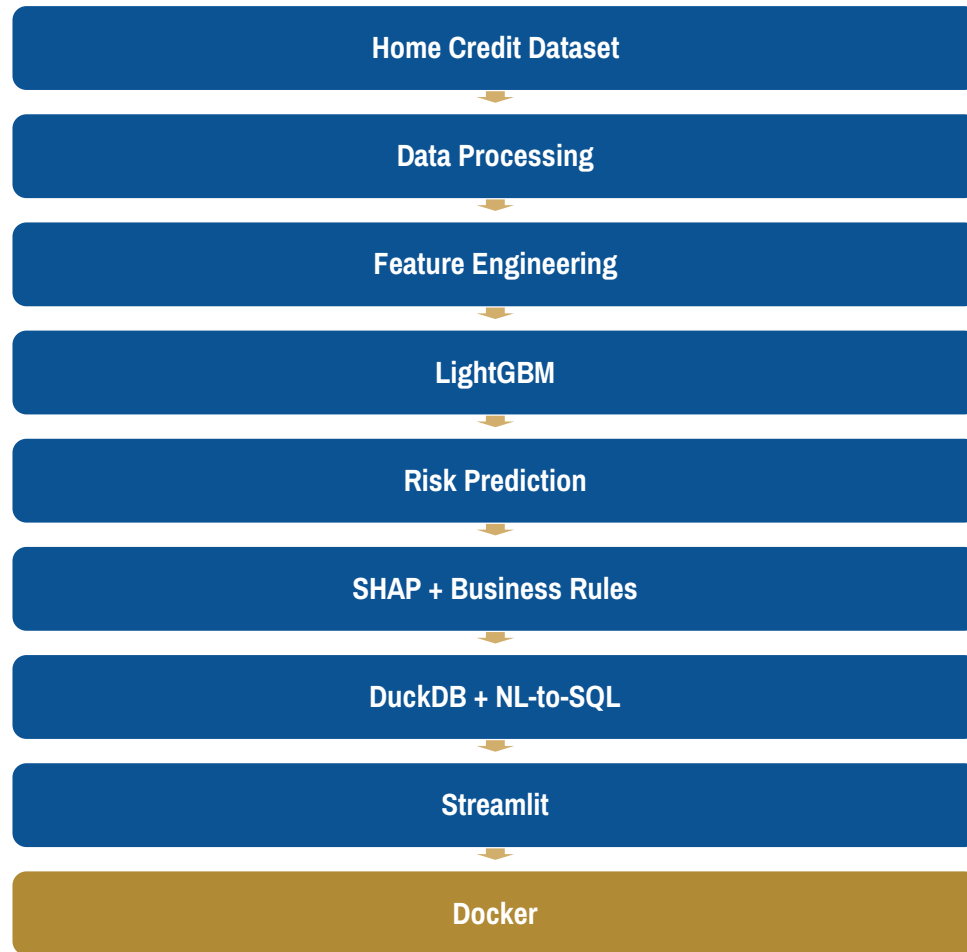
The table below shows the historical default rates observed in the dataset, segmented by education type:

Education Level	Applicants	Defaults	Default Rate
Lower secondary	5,816	417	10.93%
Secondary / secondary special	218,391	19,524	8.94%
Incomplete higher	10,277	872	8.49%
Higher education	74,863	4,009	5.36%
Academic degree	164	3	1.83%

Ask a question (e.g., "What is the default rate by education type?")

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Architecture & Deployment



Evaluator Deployment

```
docker compose up --build
```

- Fully containerized end-to-end pipeline
- Single-command setup for evaluators
- Streamlit front-end serves all platform modules

Conclusion & Key Takeaways

An end-to-end, explainable credit risk intelligence platform combining Data Engineering, Machine Learning, Explainable AI, LLM-powered analytics and containerized deployment.

8.07%

Portfolio Default Rate

0.7762

ROC-AUC

100%

Dockerized Application